# Countermovement Jump Analysis

# Data Overview

**Force Curve (Time Series)**

- Measures **vertical ground reaction force** over time
- **Normalized to 100 time points** to remove duration as a confounding factor
- **Normalized by body weight** to account for inter-athlete variance
- Serves as **input for Autoencoder and VAE models**

**Athlete Metadata**

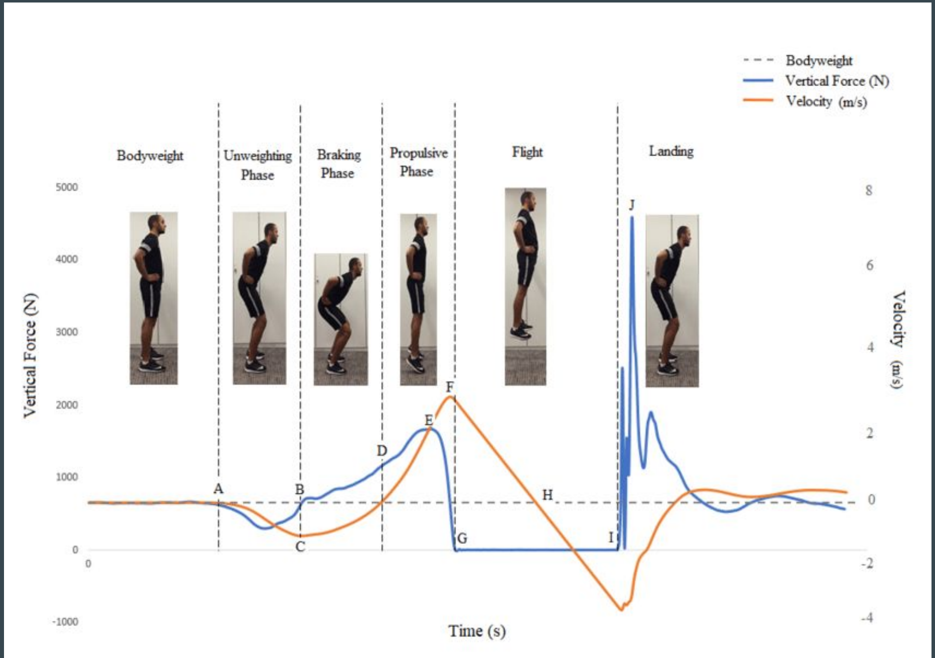
- **Athlete ID:** Unique identifier for each participant
- **Sport (Team):** Type of sport (e.g., Volleyball, Acrobatics)
- **Jump Duration:** Total time from start to landing; used to aid normalization

**Jump Performance**

- **Jump Height:** Measured (in inches) from flight time
  - Used to compare **high vs. low performers**

# What Is a Countermovement Jump?

**Definition:**
A CMJ is a vertical jump starting from an upright standing position, where the athlete first dips down (the "countermovement") before jumping upward as high as possible with their hands on their hips.



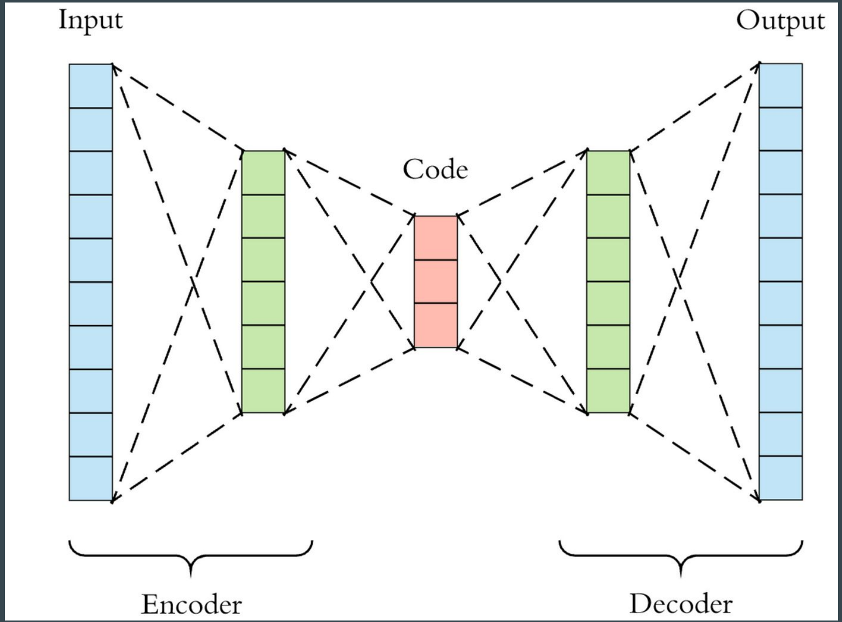

**Benefits:**

- Method of testing athlete performance
- Useful for measuring athlete's explosive lower body power
- Commonly used in strength & conditioning and sports science research
- Non-invasive and quick to administer

# Q1: Are Athletes Improving in Consistency?

# What is an Autoencoder?

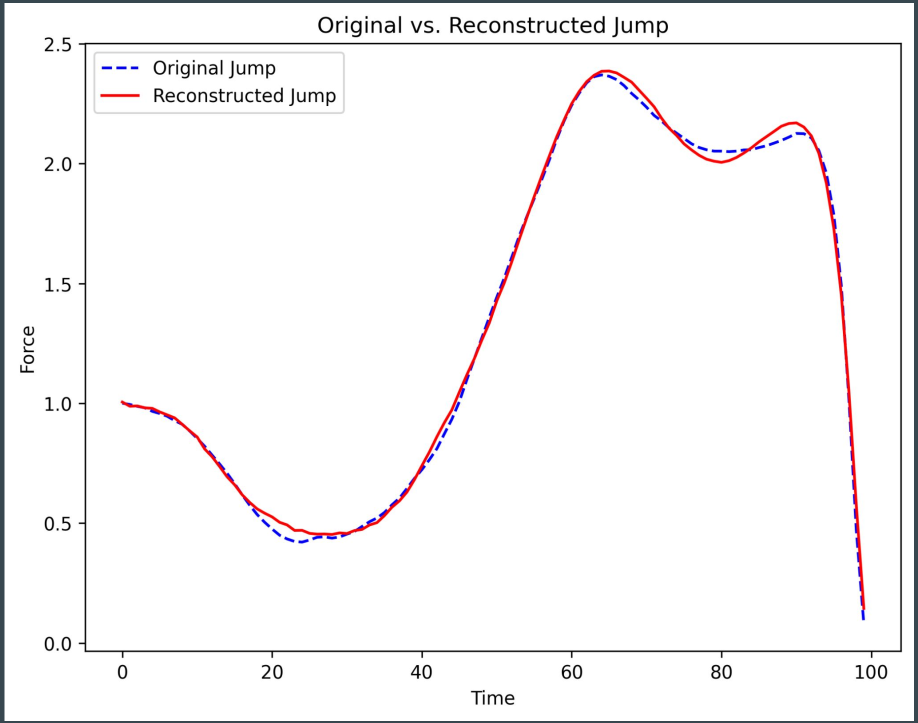
- A type of neural network used to **compress** and **reconstruct** data
    - **Compression**: Reduces data into lower-dimensional representation, called the **latent space** (similar to image compression, extracting most essential features)
    - **Reconstruction**: It then reconstructs the original data from this simplified version, ignoring unnecessary details while preserving the key structure
- In the context of jump time series, the autoencoder extracts the essential patterns that define the jump shape, while ignoring noise or insignificant variations.



# Example of Original vs. Reconstructed Jump

**Original vs. Reconstructed Jump**

- The original jump is represented as a **100-point force-time curve**.
- It's compressed into a **16-dimensional latent vector** using the encoder.
- The decoder then reconstructs the jump back to 100 points, capturing its **essential shape and strategy**.

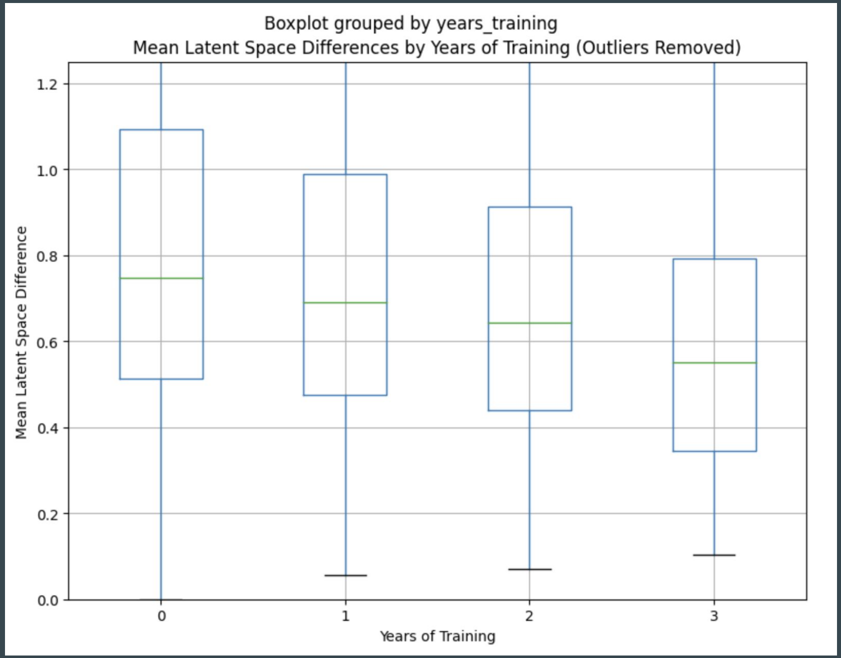


# Are Athletes Improving in Consistency?

- **Year-over-Year Trend:** Average Euclidean distances between reps and the athlete’s yearly mean latent vector **decrease over time**.

- **Interpretation:** This indicates that athletes’ movement patterns are becoming **more consistent** with each year of training.

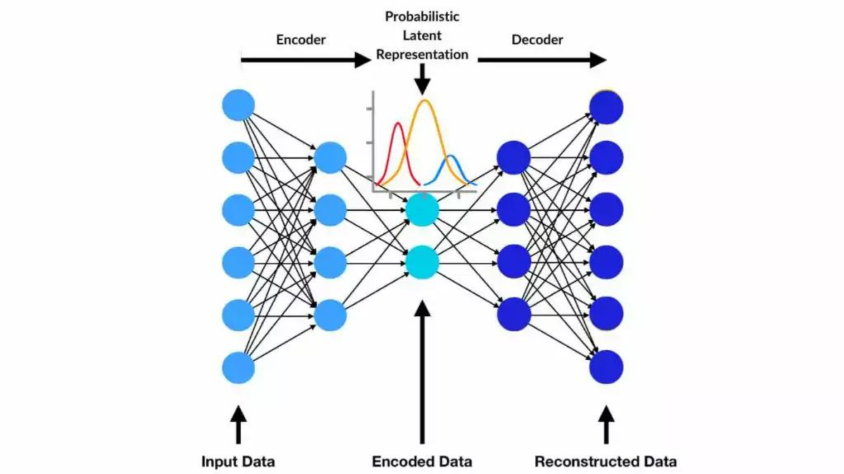
- **Conclusion:** Lower latent space variation suggests **refinement in technique** and **greater neuromuscular control** as athletes progress.



# Q2: Do Higher Jumps Share a Common Shape?

# What is a Variational Autoencoder (VAE)?

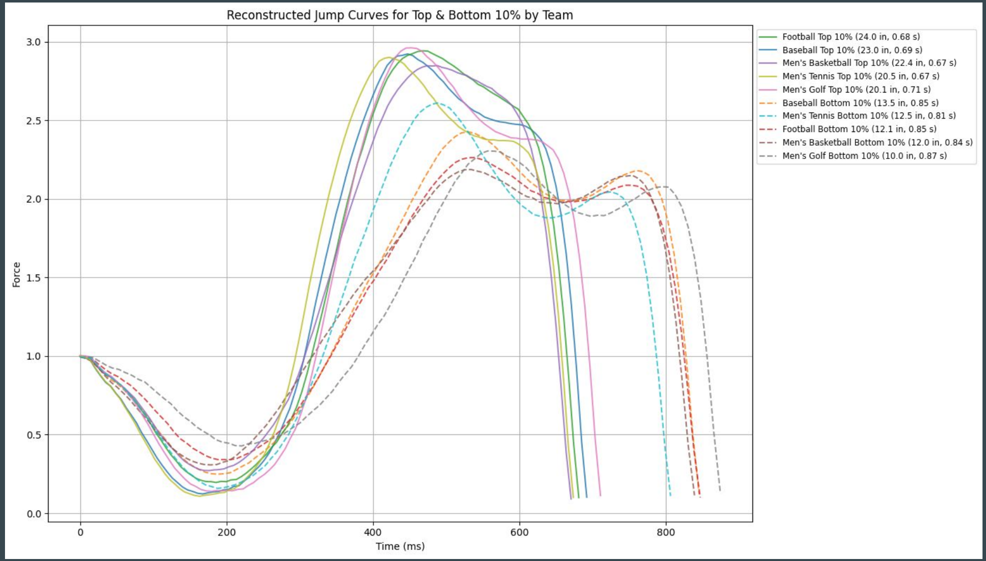
- Type of neural network used to **compress** and **reconstruct** data
  - **Compression**: Rather than encoding each input into a fixed point, VAE maps it to a  distribution (defined by mean and variance) in the latent space, meaning a jump can be represented by a region in the latent space, rather than an exact location.
  - **Reconstruction**: The model samples from this distribution to reconstruct original data, allowing VAE to learn more robust and meaningful latent spaces, which is good for recognizing patterns in shape.
- **Importance**: Similar inputs get similar distributions (i.e. similar jumps in latent space distributed near each other). Furthermore, VAE's better at generalizing important features.



# Do Higher Jumps Share a Common Shape?

## Jump Strategy Differences by Performance Level

- **Method:**
  - Computed average latent vectors per sport and per performance group (Top 10% vs. Bottom 10% by predicted jump height)
  - Used average jump durations and renormalized time to compare reconstructed force-time curves.
- **Key Finding:**
  - **Top performers** exhibit a **steeper braking and propulsive phase** with **higher relative peak force**.
  - This suggests a **more explosive and efficient jump strategy** compared to lower-performing athletes.



# Key Takeaways

- **Athletes are improving in consistency**
  - Latent space differences shrink over time, suggesting more repeatable jump patterns with training.
- **Higher jumps share a common structure**
  - Top-performing athletes exhibit distinct patterns in the latent space, reflected in their reconstructed jump shapes.
- **VAE captures meaningful biomechanical differences**
  - The model encodes differences in jump strategy and efficiency beyond just height.
- **Latent space enables performance prediction and comparison**
  - Jump height and movement quality can be inferred from position in the latent space.
- **Opportunity to prescribe performance improvements**
  - By analyzing low-performer latent vectors relative to high performers, targeted improvements can be suggested.